DATE SORTING 1 7 8 5 3 6 4 2

a) Selection Sort

i=0: min=1

1 7 8 5 3 6 4 2

i=1: min=2, 2⇆7

1 2 8 5 3 6 4 7

i=2: min=3, 8⇆3

1 2 3 5 8 6 4 7

i=3: m=4 5⇆4

1 2 3 4 8 6 5 7

i=4: min=5, 8⇆5

1 2 3 4 5 6 8 7

i=5 min=6

1 2 3 4 5 6 8 7

i=6: min=7 8⇆7

1 2 3 4 5 6 7 8

b) Insertion Sort

初始: 1 7 8 5 3 6 4 2

- i=1, key=7:

1 7 8 5 3 6 4 2

- i=2, key=8:

1 7 8 5 3 6 4 2

- i=3, key=5:

shift 8 → 1 7 8 8 3 6 4 2

shift 7 → 1 7 7 8 3 6 4 2

place 5 → 1 5 7 8 3 6 4 2

- i=4, key=3:

shift 8 → 1 5 7 8 8 6 4 2

shift 7 → 1 5 7 7 8 6 4 2

shift 5 → 1 5 5 7 8 6 4 2

place 3 → 1 3 5 7 8 6 4 2

(b) continue

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- i=5, key=6:

shift 8 → 13578842

shift 7 → 13577842

place 6 → 13567842

- i=6, key=4:

shift 8 → 13567882

shift 7 → 13567782

shift 6 → 13566782

shift 5 → 13556782

place 4 → 13456782

- i=7, key=2:

shift 8 → 13456788

shift 7 → 13456778

shift 6 → 13456678

shift 5 → 13455678

shift 4 → 13445678

shift 3 → 13345678

place 2 → 12345678

(d) Quick Sort 1 7 8 5 3 6 4 2

pivot=1 → [1, 7 8 5 3 6 4 2]

for [7 8 5 3 6 4 2]

pivot=7 → [5 3 6 4 2, 7, 8]

for [5 3 6 4 2].

pivot=5 → [3 4 2, 5, 6]

for [3 4 2]

pivot = 3 → [2, 3, 4]

so we got [1 2 3 4 5 6 7 8]

(c) Merge Sort 1 7 8 5 3 6 4 2

split [1 7 8 5 3 6 4 2]

→ [1 7 8 5] [3 6 4 2]

→ [1 7] [8 5] [3 6] [4 2]

→ [1] [7] [8] [5] [3] [6] [4] [2]

merge:

[1 7] [5 8] [3 6] [2 4]

[1 5 7 8] [2 3 4 6]

[1 2 3 4 5 6 7 8]

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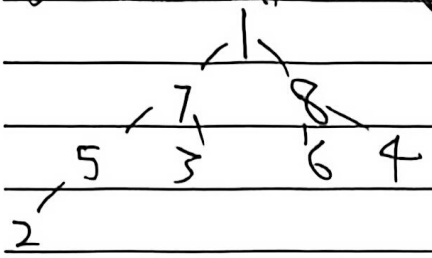
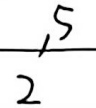
e. Heap Sort 1 7 8 5 3 6 4 2

1) Build Max Heap

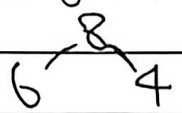


heapify index 3



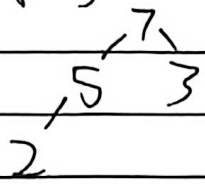
5 > 2, do nothing

heapify index 2



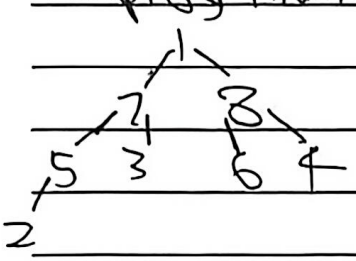
, do nothing

heapify index 1



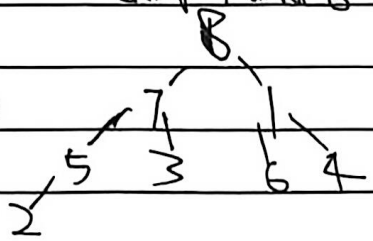
do nothing

heapify index 0

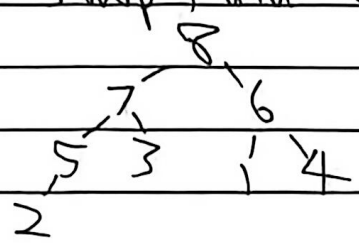


⇒

swap 1 and 8



swap 1 and 6



so we get 8, 7, 6, 5, 3, 1, 4, 2

Now first turn

swap index 0 and index 7 ⇒ 2, 7, 6, 5, 3, 1, 4, 8

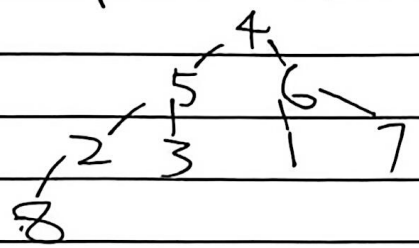
compare 2, 7, 6 ⇒ we get ⇒ 7, 2, 6, 5, 3, 1, 4, 8

continue to heapify index 1

⇒ compare 2, 5, 3, we get ⇒ 7, 5, 6, 2, 3, 1, 4, 8

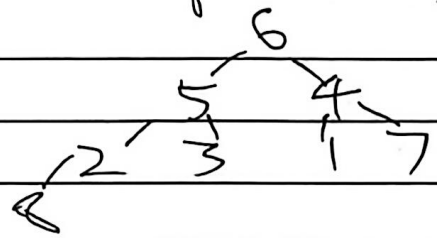
Now second turn

swap index (0) and index (6)



⇒

compare 4, 5, 6



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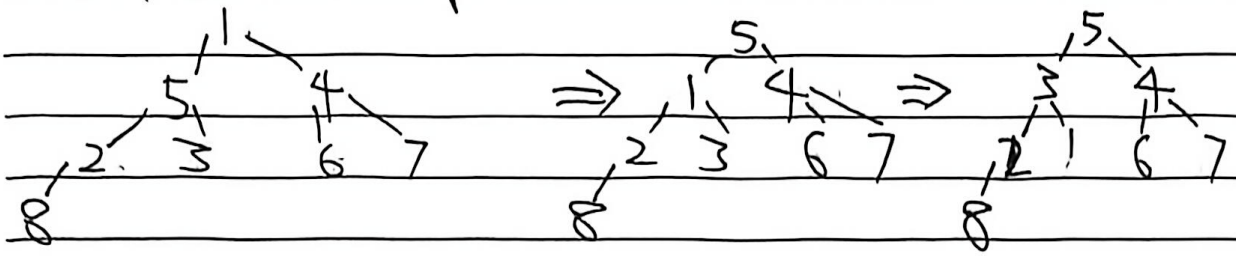
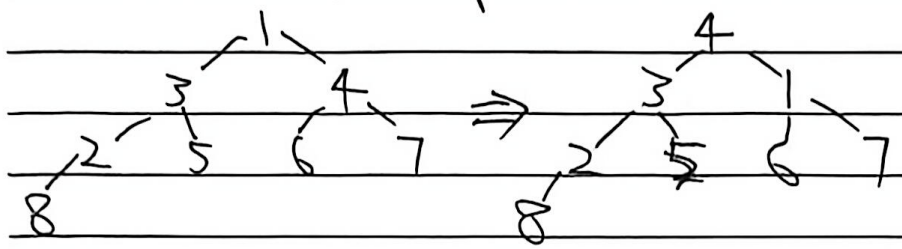
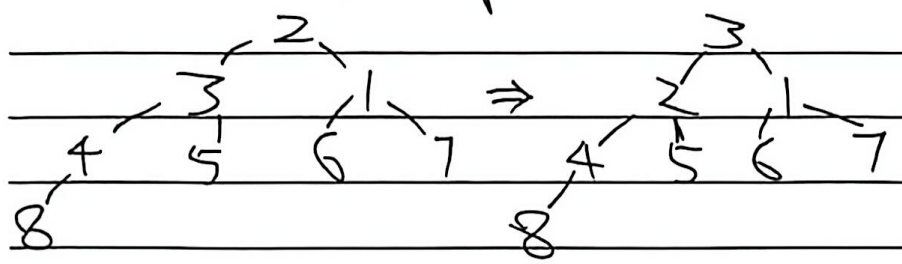
Third turn, swap index 0 and 5



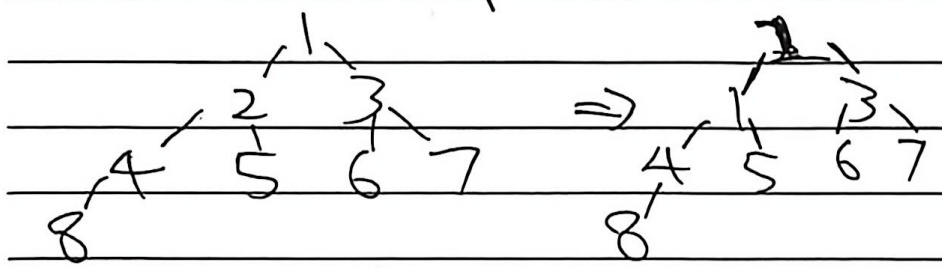
Fourth turn, swap index 0 and 4



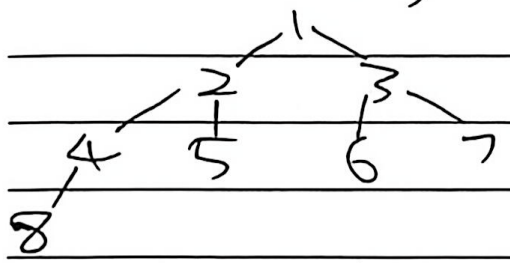
Fifth turn, swap index 0 and 3



Sixth turn swap index 0 and 2



seventh round, swap index 0 and 1



Final result is 1 2 3 4 5 6 7 8